Static Network Adjustment vs NRTK 24-Hour Solution

Elkhorn, Fresno State and SJER, 24-hour occupations, September 26–27 and October 17–18, 2026

Table 1. Site #1, Elkhorn, CA Zone 4 (0404)

Solution	Static Data			NRTK 24-hr Mean			Difference from NRTK 24-hr Mean		
	Northing (m)	Easting (m)	Elevation (m)	Northing (m)	Easting (m)	Elevation (m)	ΔN (m)	ΔE (m)	ΔElev (m)
GPS	628892.606	1929408.997	74.407	628892.607	1929409.001	74.392	-0.001	-0.004	0.015
GREC	628877.832	1929405.540	74.138	628877.834	1929405.541	74.123	-0.002	-0.001	0.015
GEC	628885.733	1929408.567	74.367	628885.728	1929408.571	74.337	0.005	-0.004	0.030
GRE	628895.834	1929406.880	74.612	628895.834	1929406.884	74.598	0.000	-0.004	0.014
GRC	628891.694	1929405.211	74.601	628891.695	1929405.215	74.588	-0.001	-0.004	0.013

Static: TBC network adjustment, September 26–27, 2026. Control: KDL1, LEMA, MULN, RAPT, RBRU (3D). Δ = static minus NRTK 24-hr mean.

Table 2. Site #2, Fresno State, CA Zone 4 (0404)

Solution	Static Data			NRTK 24-hr Mean			Difference from NRTK 24-hr Mean		
	Northing (m)	Easting (m)	Elevation (m)	Northing (m)	Easting (m)	Elevation (m)	ΔN (m)	ΔE (m)	ΔElev (m)
GPS	665103.935	1933526.493	105.608	665103.946	1933526.499	105.536	-0.011	-0.006	0.072
GREC	665094.533	1933526.323	105.515	665094.546	1933526.318	105.447	-0.013	0.005	0.068
GEC	665088.399	1933526.514	105.738	665088.415	1933526.518	105.677	-0.016	-0.004	0.061
GRE	665100.739	1933526.617	105.665	665100.750	1933526.624	105.613	-0.011	-0.007	0.052
GRC	665106.789	1933526.634	105.650	665106.801	1933526.642	105.586	-0.012	-0.008	0.064

Static: 24-hour static, October 17–18, 2026, provided in meters. Control was not provided for this station. NRTK data from 3:55 to 4:35 AM on 10/18 is not used (erratic positions on every receiver), so the NRTK 24-hr mean leaves it out. Δ = static minus NRTK 24-hr mean.

Table 3. Site #3, SJER, CA Zone 3 (0403)

Solution	Static Data			NRTK 24-hr Mean			Difference from NRTK 24-hr Mean		
	Northing (m)	Easting (m)	Elevation (m)	Northing (m)	Easting (m)	Elevation (m)	ΔN (m)	ΔE (m)	ΔElev (m)
GPS	566648.168	2066861.168	356.842	566648.170	2066861.165	356.811	-0.002	0.003	0.031
GREC	566642.163	2066857.106	358.504	566642.161	2066857.103	358.469	0.002	0.003	0.035
GEC	566637.724	2066863.061	358.261	566637.727	2066863.062	358.236	-0.003	-0.001	0.025
GRE	566632.613	2066852.112	357.638	566632.614	2066852.109	357.618	-0.001	0.003	0.020
GRC	566653.232	2066859.516	356.423	566653.231	2066859.515	356.390	0.001	0.001	0.033

Static: TBC network adjustment, September 26–27, 2026. Control: CHOW, DONO, KDL1, LEMA, MULN (horizontal), DOND (3D). Δ = static minus NRTK 24-hr mean.

Table 4. Summary by station

Station	Mean ΔN (m)	Mean ΔE (m)	Mean ΔElev (m)	Horizontal Δ, range (m)	ΔElev, range (m)
Site #1, Elkhorn	0.000	-0.003	0.017	0.002 – 0.006	0.013 – 0.030
Site #2, Fresno State	-0.013	-0.004	0.063	0.013 – 0.016	0.052 – 0.072
Site #3, SJER	-0.001	0.002	0.029	0.001 – 0.004	0.020 – 0.035

Means and ranges over the five solutions at each station. Horizontal $\Delta = \sqrt{(\Delta N^2 + \Delta E^2)}$.

Table 5. Reference stations

Site	NAD83 Latitude	NAD83 Longitude	NAD83 Height (m)	Adjustment Date	Approximate Epoch
CHOW	N 37° 05' 05.45605"	W 120° 12' 37.78225"	45.831	07/14/2026	2026.5329
DOND	N 37° 15' 51.30652"	W 119° 41' 24.40220"	666.451	07/14/2026	2026.5329
DONO	N 36° 41' 32.97972"	W 119° 01' 04.66689"	1188.501	07/14/2026	2026.5329
KDL1	N 36° 19' 10.66799"	W 119° 33' 23.29292"	46.276	07/14/2026	2026.5329
LEMA	N 36° 17' 31.28393"	W 119° 46' 56.55245"	33.182	07/14/2026	2026.5329
MULN	N 36° 33' 54.29673"	W 120° 05' 29.74980"	25.087	07/14/2026	2026.5329
RAPT	N 36° 39' 53.71288"	W 119° 26' 55.28936"	89.553	07/14/2026	2026.5329
RBRU	N 36° 45' 32.84611"	W 119° 49' 43.15487"	59.822	07/14/2026	2026.5329